The Four Fundamental Forces under the Universal Quantum Foam Hypothesis

Ukshin Q. Rexhepi
Tübingen, Germany
ukshin.rexhepi@gmail.com

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Abstract

The Universal Quantum Foam Hypothesis (UQSH) describes the four known fundamental forces not as ontologically separate interactions, but as scale-dependent modes of organization of a unified field. The local degree of organization of this field is characterized by a dimensionless parameter $Qu(x)$, which describes the ratio between structural self-binding and freedom of reorganization. Strong, electromagnetic, weak, and gravitational phenomena appear within this framework as different response modes of the same field: as local saturation binding, structuring stability order, non-persistent restructuring dynamics, and large-scale geometric persistence. The observed hierarchy of coupling strengths arises qualitatively from the scale dependence of this degree of organization. A consistent field-mechanical framework is thus proposed that interprets the interactions without introducing ontologically separate types of forces and understands gravitation as the long-range mode of incompletely bound field deformations.

Part I – Illustrative Scale Mechanics of Interactions

1 A single field reality

Within the framework of the UQSH, there does not exist a multiplicity of mutually separate forces. There is only a continuous field that organizes itself in different forms. What we refer to in physics as the strong interaction, electromagnetic force, weak interaction, or gravitation are not independent operative principles, but different manifestations of the same field reality. Depending on how strongly the field is locally saturated, interwoven, or freely reorganizable, it exhibits different behaviors. The difference therefore does not lie in the nature of a particular force, but in the respective degree of organization of the field and in the scale at which we observe it.

2 What we call forces

In this picture, a force is not an attracting or repelling entity. It is the manner in which a field deformation propagates into neighboring regions and continues there. Every interaction ultimately describes how a local disturbance of the field reorganizes itself further. If such a deformation is immediately locally bound again and transferred into stable patterns, it appears intense but is restricted to very small distances. If, on the other hand, it can continue across larger regions without being immediately captured, it appears weaker but extends further into space. Strength and range are therefore not independent properties. They are two expressions of the same field-mechanical situation and arise from the organizational state of the field itself.

3 The atom as a local field density

In this view, an atom is not a collection of small, mutually separate particles. It is a highly organized condensation of the field, a self-contained state of organization. At its center lies the nucleus. It represents a highly saturated state in which the field is extremely densely interwoven and structurally locked. This inner region is not empty space containing point-like objects, but a zone of maximal binding in which reorganization is possible only within very narrow patterns.

Surrounding this nucleus there exists a less saturated, yet nevertheless stably ordered region of the field. This is not a loose collection of individual electrons orbiting the nucleus like small planets. Rather, it is a stabilized field structure that can exist only because a specific geometric order is maintained. Here the field organizes itself in such a way that a balance arises between maximal saturation at the center and freer reorganization in the outer region. The atom is therefore not a construction set of particles, but a multilayered field configuration with graded organizational density.

4 Proton and neutron as different nuclear configurations

In this picture, protons and neutrons are not solid little spheres pressed together within the nucleus. They are different configurations of the same highly saturated field state. Both arise from extremely condensed field organization, yet they differ in their internal locking.

The proton represents a stable, independently persistent nuclear configuration. Its field structure is organized in such a way that it can maintain its saturation from within itself. It possesses a structural self-binding that allows it to exist even outside more complex nuclear associations. In this sense, the proton is a self-sustaining high-saturation form of the field.

The neutron, by contrast, is not a fully independent structure. It does not persist in isolation, but remains stable only within a composite structure. Its internal organization is such that it depends on a complementary environment. It may be conceived as a connecting field form that compensates structural imbalances within highly saturated nuclear states. It contributes to the internal order of the nucleus by distributing tension differences and stabilizing the overall geometry.

That free neutrons decay is, within this framework, not a random effect but a structural indication. Their configuration is not self-sustaining. Outside a stabilizing environment, their field state reorganizes into a more persistent form. The neutron is therefore not an isolated fundamental unit, but an integrated component of the strong interaction that unfolds its stability only in interplay with protons.

5 The strong interaction as an internal saturation bond

In this understanding, the strong interaction is not an additional mechanism that exists alongside other forces. It is the direct expression of extremely high local field saturation. Where the field is almost completely interwoven within itself, it possesses hardly any remaining freedom for reorganization. Any new deformation that enters such a region is immediately absorbed, bound, and integrated into the existing structure.

In a highly saturated nuclear regime, this gives rise to an enormous stabilizing force. Not because something actively pulls, but because the field immediately restores any deviation. The existing organization allows no free propagation. Deformations remain local, are captured, and incorporated into the existing geometry. It is precisely from this that the intensity of this interaction arises.

Its spatial limitation is therefore not a weakness, but a direct consequence of its nature. A field state that is so densely organized cannot develop an extended geometric signature. It stabilizes itself internally, but it does not project a long-range structure outward. The strong interaction thus appears short-ranged because, in this regime, the field possesses almost no freedom of reorganization. Its strength arises from maximal binding, and its limitation from precisely this bound structure.

6 Electromagnetic order as a stability condition

In this picture, the electromagnetic interaction occupies an intermediate position. It is neither as extremely saturated as the strong interaction nor as free and large-scale as gravitational geometry. It arises in a field regime of intermediate organization, in which sufficient binding is still present to ensure stability, while at the same time enough freedom of reorganization exists to build structure beyond the nucleus.

Its effect is structuring. It generates the geometric condition under which field condensations can expand without immediately collapsing back into themselves. While the strong interaction stabilizes the nucleus locally, the electromagnetic interaction enables the formation of a surrounding shell. This shell is not a loose accessory, but a stabilized field structure that can exist only if a precise geometric balance is maintained.

This becomes particularly evident in an atom. Without a compensating structure, the highly saturated nucleus would either remain isolated or transition into further local condensations. Only electromagnetic order creates a region of intermediate saturation in which stable extension becomes possible. It is the prerequisite for the formation of more complex patterns, for atoms to interact with one another, for molecules to arise, and for larger structures to be built.

One may therefore say that the electromagnetic interaction organizes the balance between binding and extension. It prevents both the immediate collapse into pure nuclear condensation and the complete dispersion into unstructured field freedom. Within this regime of intermediate saturation, the possibility of diversity, chemistry, and macroscopic structure arises.

7 The weak interaction as non-persistent field reorganization

In this picture, the weak interaction manifests where field configurations change their internal structure. It is not a permanently binding force in the sense of stable geometry, but rather a dynamics of reorganization. Whenever a highly saturated state must readjust its internal balance, this form of field transformation appears.

When a neutron transitions into a proton or a nuclear configuration readjusts itself, no additional force becomes active. Rather, the existing field configuration reorganizes. A previously integrated structure loses its stability within the given context and transitions into a new, better adapted configuration. The weak interaction describes this transitional state, not a permanent binding mechanism.

One may imagine this dynamics as smaller tension fields that arise within a larger organizational domain. They build up, modify the local structure, and dissolve again by being integrated into a more stable overall order. On the atomic scale, these processes appear punctual and rare. On the larger scale of a field regime, however, they are an expression of a continuous internal process of adjustment.

In the scale analogy, this corresponds to those smaller tension zones within a galaxy that form, dissolve again, and contribute their energy to the larger gravitational pattern. They are not the supporting geometry of the system, but they modify its internal distribution. The weak interaction is therefore not an independent principle alongside the other forces, but the visible form of structural readjustment within highly saturated field states.

8 The atom compared to the galaxy

If we consider the atom as a highly saturated field structure, the same logic of organization can be transferred to larger scales. A galaxy likewise possesses a dense center in which field organization is particularly concentrated. There, stable and long-lived structures arise, comparable to the nucleus of an atom. Around this center, however, extends a vast region whose dynamics cannot be explained solely by the visible stars and gas clouds.

In established astronomy, this context is referred to as dark matter. One observes an additional gravitational effect that cannot be directly associated with luminous matter. Within the framework of the UQSH, no new substance is required for this. What becomes effective here is the large-scale persistence of a field deformation that is not immediately locally bound. Whereas in highly saturated regions every new structure is rapidly integrated and stabilized, in less saturated regimes extensive tension landscapes can form that persist over large distances.

This persistent geometry acts gravitationally without being identifiable as an independent object. It is not additional matter, but an expression of the field organization itself. Just as an atom consists not only of its nucleus, but also of an extended shell structure that determines its overall effect, a galaxy likewise consists not only of its visible components. Its gravitational signature encompasses the entire tension landscape of the field in which it is embedded.

9 Mini-dark matter as a scale analogy

If every highly saturated field structure, in addition to its local binding, also generates a more extensive geometric environment, it is natural to understand this principle as scale-transcending. On the galactic level, we speak of dark matter because we observe an additional gravitational effect that cannot be directly derived from the visible matter. We see the tension landscape, but not its complete structural origin.

On the atomic level, this question does not arise in the same form. Our measurement methods there respond almost exclusively to bound interactions. They are designed to register charge, momentum, energy transfer, or decay processes. A possible scale-relative, cumulative field persistence would be absorbed into these known interactions. It would not appear as an isolated independent effect, but as part of the overall structure of the bound system.

The term mini dark matter is therefore not a new physical entity and not a claim of a hidden substance within the atom. It serves exclusively as an illustrative analogy. It is intended to clarify that the principle of a not fully locally bound field persistence may operate on all scales, even where we cannot identify it separately due to our mode of observation. It is not a proof, but a scale-mechanical illustration of a unified field principle.

10 Why gravity is the long-range mode

Within this framework, gravitation is not an additional force alongside other forces. It is the long-range mode of the same field mechanics from which all other interactions also emerge. It arises where a field deformation is not immediately locally bound and integrated into a compact structure, but instead remains as a large-scale geometric modification.

While in highly saturated regimes every new deformation is immediately nested and absorbed, in less saturated regions it can continue to act as a persistent tension landscape. It is precisely this persistent geometry that we experience as gravitation. It does not act through pulling or pushing, but by altering the organizational state of the field itself and thereby influencing the possible motions of all other structures.

Therefore, gravitation is universal. Every field organization, regardless of scale, carries such a geometric signature. And this is why it appears weak. Not because it is less real or less fundamental, but because it is the portion of a deformation that is not locally bound and remains as an extended structure. Gravitation is thus not a special case, but the most persistent expression of a continuously reorganizing field.

Part II – Field-mechanical formulation of the scale-dependent interactions

11 Fundamental field structure and organizational parameters

Within the framework of the UQSH, there exists a continuous field $\mathcal{F}$ whose local states of organization give rise to all observable phenomena.

To describe the local degree of organization, the dimensionless parameter

$$Qu(x)$$

is introduced.

Definition:

$$Qu(x) := \text{local Organizational or Saturation level from } \mathcal{F}$$

on the position x.

The parameter $Qu(x)$ measures neither energy nor mass nor temperature. It describes the degree of structural self-binding of the field relative to free reorganization. A high value means that the field is strongly nested and locally bound. A low value means that the field possesses a high degree of freedom for reorganization.

In this way, no additional material parameter is introduced, but rather a measure of the extent to which the field at a given location is compelled to preserve existing organization instead of freely reshaping itself.

12 Effective coupling as a function of field saturation

A local field deformation $\delta\mathcal{F}$ couples with an effective coupling strength

$$g_{\text{eff}}(x) = g(Qu(x)),$$

where the following holds

$$\frac{dg_{\text{eff}}}{dQu} > 0.$$

This relation means: The higher the degree of organization $Qu(x)$, the stronger the local coupling.

In intuitive terms, this means: In highly saturated field regimes, every new deformation is immediately absorbed, integrated, and embedded into the existing structure. As a result, the interaction appears intense, but it remains locally confined.

In weakly saturated regions, by contrast, the immediate binding is weaker. The deformation is not captured at once and can propagate further. Strength is therefore not an independent law, but the result of the local state of organization.

13 Range as an integral effect of local reorganization

To describe the range, a local reorganization damping is introduced:

$$\kappa(x) = \kappa(Qu(x))\,,$$

with

$$\frac{d\kappa}{dQu} > 0.$$

The effective range along a path γ is obtained approximately as

$$R(\gamma) \sim \left(\int_\gamma \kappa(x)\, ds \right)^{-1}.$$

This relation describes that range is not an independent parameter of a force. It arises from the integration of local field conditions.

If $\kappa(x)$ is large, a deformation is quickly bound locally, and the range remains small. If $\kappa(x)$ is small, the deformation can continue to propagate, and the range increases.

Range is thus the result of how quickly the field converts a deformation back into stable organization.

14 Gravitation as the persistence mode of field deformation

In addition to the local coupling, a persistence component $\Pi(x)$ is introduced:

$$\Pi(x) = \Pi(Qu(x)), \qquad 0 \leq \Pi(x) \leq 1,$$

with

$$\frac{d\Pi}{dQu} < 0.$$

$\Pi(x)$ describes the portion of a field deformation that is not immediately locally bound, but remains as a geometric structure.

In highly saturated regimes, $\Pi(x)$ is small, since almost every deformation is immediately integrated. In weakly saturated regimes, a larger portion can remain as a geometric signature.

Gravitation corresponds precisely to this persistence mode. It is not an additional force, but the large-scale geometric consequence of field deformations that are not fully bound.

It is weak because it does not appear as an intensive local binding. It is universal because every stable field organization leaves behind a geometric trace.

15 Scale regimes of the interactions

Depending on the magnitude of $Qu(x)$, different regimes of interaction arise:

$$Qu(x) \gg 1 \quad \Rightarrow \quad \text{strong local binding, low range}$$
$$Qu(x) \sim 1 \quad \Rightarrow \quad \text{stabilizing ordering dynamics}$$

$$Qu(x) \ll 1 \quad \Rightarrow \quad \text{geometric persistence, large range}$$

In highly saturated regimes, bound states dominate, as they occur in the atomic nucleus. In intermediate regimes, structuring stability arises, as is characteristic of electromagnetically ordered systems. In weakly saturated regimes, the field deformation remains as large-scale geometry, which appears as gravitation.

Thus, the strong, electromagnetic, and gravitational regimes are not introduced as separate ontological forces, but are understood as scale-dependent modes of organization of a unified field.

16 The weak interaction as a restructuring mode

In addition to local binding and geometric persistence, there exists a third field-mechanical mode: structural reorganization.

This mode describes situations in which an existing field configuration changes its internal structure without giving rise to a new lasting geometric persistence.

Formally, this can be written as a transition dynamics:

$$\partial_t Qu(x) \neq 0 \quad \text{with simultaneously finite} \quad g_{\text{eff}}(x).$$

This means: The degree of organization changes locally without building up a large-scale persistence.

Within this framework, the weak interaction does not correspond to a binding mechanism, but to an internal reorganization dynamics in which field configurations readjust their structural balance.

Examples are structural transitions within highly saturated nuclear regimes. In conventional physics, this is described as a transition of a neutron into a proton. However, this language presupposes that there are two clearly separated particles that change their identity.

Within the framework of the UQSH, a neutron is not an isolated object. It is a specific form of integration within a highly saturated nuclear field state. While the proton represents an independently persistent saturation structure, the neutron is a configuration form that remains stable only within a structural composite. It corresponds to a specific degree of binding or a state of equilibrium within the strong field organization.

When such a nuclear state reorganizes, it is not one particle that changes into another, but the internal saturation geometry of the entire field composite that is altered. What is classically described as a change of identity is here a reordering of the degree of organization within the same underlying field structure.

The weak interaction therefore does not denote an exchange of substances and no transformation of objects. It describes a local restructuring dynamics in which the degree of binding within a highly saturated field regime readjusts. The dynamics is transforming, not persisting. No new long-range structure arises, but rather a changed internal balance of the same field system.

It changes configurations instead of generating stable geometric long-range structures.

17 Protons and neutrons as highly saturated nuclear modes

In highly saturated regimes with

$$Qu(x) \gg 1$$

stable nuclear configurations arise.

A proton represents an independently persistent saturation structure. Its field organization is self-supporting, which allows it to remain stable in isolation.

A neutron, by contrast, is not a fully independent saturation structure. It exists permanently only within a composite.

Formally, this means that its stability depends on a surrounding saturation gradient:

$$\text{stability}_n \propto \nabla Qu(x).$$

Free neutrons decay because outside the highly saturated composite regime they do not represent a self-supporting state of organization.

They are integrated nuclear configurations that compensate for structural imbalances within strong saturation regimes.

18 Scale analogy: atom and galaxy

The field mechanics described is not restricted to atomic scales.

A galaxy likewise possesses a highly condensed center and an extended environment of lower saturation.

Formally, one can write:

$$Qu_{\text{Center}} \gg Qu_{\text{Halo}}.$$

While strong local binding dominates at the center, a significant persistence component remains in the surrounding region.

$$\Pi(x) > 0.$$

This large-scale persistence appears as an additional gravitational effect that is not fully explained by visible bound matter.

Within the framework of the UQSH, this is not a new substance, but the geometric continuation of highly saturated field organization into less saturated regions.

19 Scale-relative field persistence and analogy to dark matter

If every highly saturated field structure generates a geometric environment, then this principle is scale-spanning.

At the galactic level, this manifests as what is conventionally referred to as dark matter.

At the atomic level, corresponding persistent components cannot be measured in isolation, since our detectors respond exclusively to bound interactions.

A possible scale-relative cumulative field persistence would be absorbed into known interactions and would not appear as a separate effect.

The term "mini dark matter" serves here solely as an analogy to illustrate the scale continuity of field persistence. It implies no new substance and no additional particle.

20 Qualitative scaling of the coupling hierarchy

A central question of any unified field description is whether the observed hierarchy of interactions can at least qualitatively be derived from the organizational parameter $Qu(x)$.

In established physics, the dimensionless coupling constants differ considerably in their magnitude. The strong interaction is of order 1, the electromagnetic one about 10^{-2}, the weak one significantly smaller, and gravitation extremely weak.

Within the framework of the UQSH, none of these quantities is introduced as a fundamentally separate constant. Instead, it is assumed that the effective coupling results from the local degree of organization:

$$g_{\text{eff}} \sim f(Qu)\,.$$

Qualitatively, the following holds:

$$Qu \gg 1 \quad \Rightarrow \quad g_{\text{eff}} \sim \mathcal{O}(1)$$

$$Qu \sim 1 \quad \Rightarrow \quad g_{\text{eff}} \ll 1$$

$$Qu \ll 1 \quad \Rightarrow \quad g_{\text{eff}} \to 0$$

In highly saturated regimes, nearly every field deformation is immediately integrated locally. The coupling appears strong and of order one. This corresponds to the strong interaction regime.

In intermediate organizational regimes, only a part of the deformation is locally bound, while another part remains capable of reorganization. The coupling is significantly smaller, yet remains structure-forming. This behavior is qualitatively consistent with the electromagnetic interaction.

In weakly saturated regimes, finally, only a minimal portion of a deformation is locally bound. The predominant portion persists as a geometric signature. The effective coupling appears extremely weak, while the range becomes large. This corresponds to the gravitational regime.

The scaling presented here does not claim a numerical derivation of the known constants. It does, however, show that a hierarchical structure of couplings can naturally arise from a scale-dependent field organization, without introducing ontologically separate types of interactions.

The coupling hierarchy thus does not appear as a coincidence of several fundamental parameters, but as an expression of different organizational regimes of the same field.

21 Unified framework of the interactions

The four known interaction regimes can thus be interpreted as different modes of organization of the same field:

$$\begin{aligned} Qu(x) \gg 1 &\Rightarrow \text{strong local binding} \\ Qu(x) \sim 1 &\Rightarrow \text{structuring stability} \\ \partial_t Qu(x) \neq 0 &\Rightarrow \text{restructuring dynamics (weak)} \\ Qu(x) \ll 1 &\Rightarrow \text{geometric persistence (gravitational)} \end{aligned}$$

Thus, the strong, electromagnetic, weak, and gravitational regimes are not introduced as separate ontological forces, but as scale-dependent response modes of a unified field.

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